Agent Operational Specification

Section 2 — Apex Insurance Group Integrated Architecture Package

Project	Apex Insurance Group Integrated Architecture and Sign-off Package
Template Section	Section 2 — Agent Operational Specifications
Learner Name	Dennis Lam
Deliverable	AI-Graded File Upload (PDF)

Section summary. The Claims Processing chain is four single-responsibility task agents in proj-apex-claims-processing (the Section 1 Claims Project). A claim PDF landing in claims-inbox triggers the chain, and the Claims Workflow Controller (Section 3) coordinates it. Each agent reads typed inputs and writes its own block of one JSON record, ClaimCaseOutput v1.0, which is also the only source of field names for the Section 3 A2A handoff. Seven threshold-based escalation conditions send claims to people. The Fraud and Risk Specialist is a bounded autonomous agent specified with the five pillars, and every grounding source for every agent sits in the Claims Project.

Part A — Claims Processing Task Agent Chain

1. Task Agent Inventory

Agent Name	Single Responsibility	Project Boundary	Grounding Source(s)	Downstream Consumer
Claim Intake / Extraction Agent (claim-intake-extraction-agent)	Extract structured data from each incoming commercial property damage claim PDF (ACORD 1 Property Loss Notice plus attachments) into the extracted_claim block, with a confidence score for every field.	proj-apex-claims-processing	• claims-inbox on stapexclaimsintake (claim PDFs, read-only) • Document Intelligence di-apex-claims-prod, custom model apex-acord1-property-loss-v3 • Foundry IQ kb-claims-intake-guide (ACORD 1 field guide, Apex extraction rules)	Claims Workflow Controller, which sends the claim to the Policy Cross-Reference Agent (or to sbq-intake-exceptions on ESC-003).
Policy Cross-Reference Agent (policy-crossref-agent)	Cross-reference the extracted claim against the internal policy database: policy in force on the date of loss, coverage status, limit, deductible and applicable exclusions (policy_cross_reference block).	proj-apex-claims-processing	• sqldb-apex-policy view vw_policy_coverage (read-only) • Foundry IQ kb-claims-policy-wordings (commercial property forms and endorsements)	Claims Workflow Controller, which sends the claim to the Fraud and Risk Specialist (or to sbq-coverage-review on ESC-004).
Fraud and Risk Specialist (fraud-risk-specialist)	Assess fraud and risk using the external public-records API and internal fraud indicators; produce fraud_risk_score and risk_classification with cited evidence (fraud_assessment block).	proj-apex-claims-processing	• Public-records API via apim-apex-integration-prod (/public-records/v1, GET only) • Foundry IQ kb-claims-fraud-indicators (SIU red-flag guidelines, scoring rubric) • sqldb-apex-policy view vw_claim_history_36m (read-only)	Routing gate in the Claims Workflow Controller: Settlement Specialist on the standard path, SIU queue sbq-siu-fraud-review on ESC-002, hold queue on ESC-005.
Settlement Specialist (settlement-specialist)	Draft the settlement letter and settlement recommendation for adjuster approval (settlement block). It drafts only; it never sends a letter or releases payment.	proj-apex-claims-processing	• Foundry IQ kb-claims-settlement-templates (approved letter templates, state disclosure clauses) • calculate_settlement tool (deterministic arithmetic)	Settlement output validation gate (rules V1 to V9), then the Letter Rendering Service, then the assigned adjuster's review queue, where a person releases the

Agent Name	Single Responsibility	Project Boundary	Grounding Source(s)	Downstream Consumer
				letter.

Agents never call each other. Every transition between them, and every escalation, is executed by the Claims Workflow Controller (`claims-workflow-controller`, specified in Section 3), which validates each agent's output block against the schema in item 4 before moving on.

2. Four-Step Task Agent Design Protocol

Protocol Step	Required Entry	Learner Response
Trigger	Reference an actual claim document ingestion event.	**Event:** Azure Event Grid Microsoft.Storage.BlobCreated on container claims-inbox of stapexclaimsintake, for a blob with content type application/pdf. In plain terms: a new commercial property damage claim PDF has been submitted through the broker portal, email intake or adjuster upload. **Handling:** the Claims Workflow Controller subscribes to the event, reads the blob metadata, creates correlation_id = {claim_id}.{workflow_run_id} and starts the chain at the Claim Intake / Extraction Agent. Duplicate events with the same claim_id and document_etag are ignored, so a claim never runs twice.
Typed Inputs	List named fields and data types.	Nine named, typed fields from the ingestion event, defined in item 3: claim_document_ref (string, URI), claim_id (string), policyholder_id (string), claim_submission_timestamp (datetime), submission_channel (enum), document_content_type (string), document_etag (string), supporting_document_refs (array of URI), correlation_id (string). The controller rejects the event before any agent runs if a required field is missing or mistyped.
JSON Output Schema	Define downstream-compatible structured output.	ClaimCaseOutput v1.0, a JSON Schema (draft 2020-12) defined in item 4. It has one record per claim, with blocks extracted_claim, pii_tokens, policy_cross_reference, fraud_assessment, redacted_claim_summary, routing and settlement. Each agent writes only its own block, and the controller validates the record after every step. The record feeds the settlement system directly, and it is the only source of field names for the Section 3 A2A handoff.
Escalation Conditions	Provide at least two specific thresholds.	Seven threshold-based conditions, listed in item 5. The two primary thresholds are **ESC-001**, claimed_amount_usd > $250,000 (Large Loss Unit approval), and **ESC-002**, fraud_risk_score ≥ 0.85 (SIU review; the Settlement Specialist is bypassed). Others cover extraction confidence below 0.80, no coverage, an external API that stays unavailable for more than 4 hours, elevated risk from 0.60 to 0.849, and a failed letter validation.

3. Typed Input Schema

Field Name	Data Type	Required?	Source	Description
claim_document_ref	string (URI)	Yes	Event Grid BlobCreated event, data.url	Blob URI of the incoming commercial property damage claim PDF (ACORD 1 Property Loss Notice plus adjuster narrative) in claims-inbox.
policyholder_id	string, pattern PH-NNNNNNNN	Yes	Blob metadata policyholder_id, written by the claims intake portal at submission	Apex internal policyholder key. Scopes the Policy Cross-Reference lookup to the insured's own policies.
claim_submission_timestamp	datetime (ISO 8601, UTC)	Yes	Event Grid eventTime	Time the claim was received. Starts the claim-handling SLA clock and the 4-hour external-verification timer (ESC-005).
claim_id	string, pattern CLM-YYYY-NNNNNNN	Yes	Blob metadata claim_id, assigned by the intake portal at first notice of loss	Business key of the claim and prefix of correlation_id.
submission_channel	string enum: BROKER_PORTAL, EMAIL_INTAKE, ADJUSTER_UPLOAD	Yes	Blob metadata submission_channel	Selects channel-specific extraction rules (for example, email-intake PDFs carry a cover-email page to skip).
document_content_type	string (must equal application/pdf)	Yes	Event Grid data.contentType	Non-PDF uploads are rejected before any agent runs.
document_etag	string	Yes	Event Grid data.eTag	Idempotency key together

Field Name	Data Type	Required?	Source	Description
				with claim_id; duplicate events for the same blob version are ignored.
supporting_document_refs	array<string (URI)>	No	Blobs under the same claims-inbox/{claim_id}/ prefix	Photos, repair estimates and invoices. Used as evidence, not as the source of truth for extraction.
correlation_id	string, {claim_id}.{workflow_run_id} (UUID v4)	Yes	Generated by the Claims Workflow Controller at trigger time	Passed to every agent call, span, queue message and A2A handoff so that compliance can rebuild the full run.

4. JSON Output Schema

Schema ClaimCaseOutput v1.0 (JSON Schema draft 2020-12). Fields marked `x-pii` are stored only in the Restricted `claim-cases` container of `cosmos-apex-claims-prod` and never cross the Fraud-to-Settlement A2A boundary; downstream consumers receive the tokenized and redacted fields instead.

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.apex-insurance.example/claims/ClaimCaseOutput/1.0",
  "title": "ClaimCaseOutput",
  "type": "object",
  "required": ["schema_version", "claim_id", "correlation_id", "workflow_status", "extracted_claim", "pii_tokens",
     "policy_cross_reference", "fraud_assessment", "redacted_claim_summary", "routing", "settlement"],
  "properties": {
    "schema_version": {"type": "string", "const": "1.0"},
    "claim_id": {"type": "string", "pattern": "^CLM-[0-9]{4}-[0-9]{7}$"},
    "correlation_id": {"type": "string", "pattern": "^CLM-[0-9]{4}-[0-9]{7}\\.[0-9a-f-]{36}$"},
    "workflow_status": {"enum": ["EXTRACTED", "POLICY_VERIFIED", "FRAUD_ASSESSED", "AWAITING_EXTERNAL_VERIFICATION", "ESCALATED",
      "SETTLEMENT_DRAFTED", "LETTER_VALIDATED"]},
    "extracted_claim": {
      "type": "object",
      "required": ["policyholder_id", "policy_number", "insured_entity_name", "property_address", "property_location_state",
     "date_of_loss", "loss_type", "loss_description", "claimed_amount_usd", "extraction_confidence_min"],
      "properties": {
        "policyholder_id": {"type": "string", "pattern": "^PH-[0-9]{8}$"},
        "policy_number": {"type": "string", "x-pii": true},
        "insured_entity_name": {"type": "string", "x-pii": true},
        "property_address": {"type": "string", "x-pii": true},
        "property_location_state": {"type": "string", "pattern": "^[A-Z]{2}$"},
        "date_of_loss": {"type": "string", "format": "date"},
        "loss_type": {"enum": ["FIRE", "WATER", "WIND_HAIL", "THEFT_VANDALISM", "COLLAPSE", "OTHER"]},
        "loss_description": {"type": "string", "x-pii": true},
        "claimed_amount_usd": {"type": "number", "minimum": 0},
        "extraction_confidence_min": {"type": "number", "minimum": 0, "maximum": 1}
      }
    },
    "pii_tokens": {
      "type": "object",
      "required": ["policyholder_token", "policy_number_token"],
      "properties": {
        "policyholder_token": {"type": "string", "pattern": "^PH-TKN-[0-9a-f]{16}$"},
        "policy_number_token": {"type": "string", "pattern": "^POL-TKN-[0-9a-f]{16}$"}
      }
    },
    "policy_cross_reference": {
      "type": "object",
      "required": ["policy_in_force_on_loss_date", "coverage_status", "coverage_limit_usd", "deductible_usd",
     "applicable_exclusions"],
      "properties": {
        "policy_in_force_on_loss_date": {"type": "boolean"},
        "coverage_status": {"enum": ["COVERED", "PARTIAL", "NOT_COVERED"]},
        "coverage_limit_usd": {"type": "number", "minimum": 0},
        "deductible_usd": {"type": "number", "minimum": 0},
        "applicable_exclusions": {"type": "array", "items": {"type": "string"}}
      }
    },
    "fraud_assessment": {
      "type": "object",
      "required": ["fraud_risk_score", "risk_classification", "fraud_signals", "public_records_check_status", "evidence_refs",
     "assessed_at"],
      "properties": {
        "fraud_risk_score": {"type": ["number", "null"], "minimum": 0, "maximum": 1},
        "risk_classification": {"enum": ["LOW", "ELEVATED", "HIGH", "INDETERMINATE"]},
        "fraud_signals": {"type": "array", "items": {"type": "string", "pattern": "^FS-[0-9]{3}$"}},
        "public_records_check_status": {"enum": ["COMPLETED", "CACHED", "UNAVAILABLE"]},
        "evidence_refs": {"type": "array", "items": {"type": "string"}},
        "assessed_at": {"type": "string", "format": "date-time"}
      }
    },
```

```
    "redacted_claim_summary": {"type": "string", "maxLength": 1500},
    "routing": {
      "type": "object",
      "required": ["route", "settlement_hold_required", "escalation_ids"],
      "properties": {
        "route": {"enum": ["SETTLEMENT", "SIU_REVIEW", "EXTERNAL_VERIFICATION_HOLD", "INTAKE_EXCEPTION", "COVERAGE_REVIEW",
      "ADJUSTER_MANUAL_REVIEW"]},
        "settlement_hold_required": {"type": "boolean"},
        "escalation_ids": {"type": "array", "items": {"type": "string", "pattern": "^ESC-[0-9]{3}$"}}
      }
    },
    "settlement": {
      "type": ["object", "null"],
      "required": ["recommended_settlement_amount_usd", "settlement_letter_ref", "letter_validation_status"],
      "properties": {
        "recommended_settlement_amount_usd": {"type": "number", "minimum": 0},
        "settlement_letter_ref": {"type": "string", "format": "uri"},
        "letter_validation_status": {"enum": ["PENDING", "PASSED", "FAILED"]}
      }
    }
  }
}
```

Who writes each block: Intake writes `extracted_claim`. The controller's non-LLM tokenization and redaction step, run straight after extraction, writes `pii_tokens` and `redacted_claim_summary`. Policy Cross-Reference writes `policy_cross_reference`, Fraud and Risk writes `fraud_assessment`, Settlement writes `settlement`, and the controller writes `routing` and `workflow_status`.

How it feeds the downstream settlement system: the Letter Rendering Service and the adjuster review queue consume `claim_id`, `pii_tokens`, `settlement.recommended_settlement_amount_usd`, `settlement.settlement_letter_ref`, `settlement.letter_validation_status` and `routing.settlement_hold_required`. No downstream system has to parse free text. `recommended_settlement_amount_usd` comes from the deterministic `calculate_settlement` tool: the lower of the covered loss and `coverage_limit_usd`, minus `deductible_usd`.

5. Escalation Conditions

Condition ID	Triggering Metric or Event	Threshold	Escalation Destination	Workflow Effect
ESC-001	Claim value: claimed_amount_usd	> **$250,000**	Large Loss Unit (Senior Commercial Property Adjuster) via sbq-large-loss-approval	Chain continues, but settlement_hold_required = true: the letter is created as DRAFT_HOLD and cannot be released without Large Loss Unit approval. Cached public-records results are not accepted; a live check is required.
ESC-002	Fraud confidence score: fraud_risk_score	**≥ 0.85** (risk_classification = HIGH)	Special Investigations Unit (SIU) via sbq-siu-fraud-review	Settlement Specialist is bypassed and no letter is drafted. The workflow is checkpointed and suspended until an SIU decision event arrives: CLEAR resumes the handoff with the reviewer recorded; CONFIRM ends at SIU referral.
ESC-003	Extraction quality: extraction_confidence_min, or a required extracted_claim field missing	< **0.80**, or at least one required field null	Claims Intake Exceptions team via sbq-intake-exceptions	Chain stops before policy cross-reference. A person corrects or confirms the fields, and the chain restarts at the Policy Cross-Reference Agent with the same correlation_id.
ESC-004	Coverage: policy_in_force_on_loss_date, coverage_status	**false**, or **NOT_COVERED**	Claims Coverage Review team (Operations) via sbq-coverage-review	Fraud and Settlement steps do not run and no settlement letter is drafted. Coverage position is handled by people inside Operations (never routed to Legal, which has no access to claims data).
ESC-005	External verification: public_records_check_status	**UNAVAILABLE for > 4 hours** after the first attempt (circuit open, no valid cache of 30 days or less)	Assigned commercial property adjuster via sbq-adjuster-review	Claim is parked as AWAITING_EXTERNAL_VERIFICATION with a checkpoint and resumes automatically when the circuit closes. After 4 hours the adjuster runs a manual records check. Settlement never runs on an INDETERMINATE assessment.
ESC-006	Elevated fraud risk: fraud_risk_score	**≥ 0.60 and < 0.85** (ELEVATED)	Assigned adjuster (approval required)	Handoff proceeds with settlement_hold_required = true. The letter stays DRAFT_HOLD until the adjuster reviews fraud_signals and approves.
ESC-007	Letter quality: letter_validation_status	**FAILED** after 1 regeneration attempt	Assigned adjuster (manual drafting) via sbq-adjuster-review	Draft is quarantined, not rendered or released. The adjuster drafts manually with the field-level validation errors attached (Section 4, FN-002).

Part B — Fraud Risk Assessment Agent Specification

6. Autonomous or Task-Agent Design Choice

Design Decision	Learner Response
Selected agent type	**Autonomous Agent** with bounded autonomy, running as one step inside the orchestrated Claims chain.
Justification	Fraud assessment is an open-ended investigation: which public-records lookups to run depends on what earlier lookups return. An entity-name mismatch calls for business registration history and liens. A policy bound weeks before the loss calls for prior-claim history. A clean first pass may need nothing more. A task agent with a fixed sequence would either over-query (cost and vendor quota) or miss signals. Autonomy is bounded by the five pillars below: read-only tools, a budget of 12 tool calls or 120 seconds, a deterministic scoring tool so the score is reproducible, and an asynchronous human checkpoint. The agent only classifies risk. It never approves, denies or pays, and routing stays with the deterministic controller.

7. Five-Pillar Fraud Risk Agent Specification

Pillar	Learner Response
Goal State	For each claim_id handed to it, a complete fraud_assessment block exists in which fraud_risk_score (0.00 to 1.00) and risk_classification are backed by cited evidence, so the Workflow Controller can route the claim deterministically. Bands: **LOW** < 0.60; **ELEVATED** 0.60 to 0.849; **HIGH** ≥ 0.85; **INDETERMINATE** when the external check is unavailable and no valid cache exists (score = null). The goal is an accurate classification, not a particular outcome: the agent is not measured on how many claims it clears.
Definition of Done	Done only when **all** of the following are true: • Five mandatory checks were run or explicitly marked unavailable: (1) insured entity registration status and name match; (2) property ownership for the loss address; (3) liens, judgments and litigation on the insured entity; (4) 36-month prior-claim history from vw_claim_history_36m; (5) review against kb-claims-fraud-indicators. • fraud_risk_score was produced by the score_fraud_signals tool (never typed by the model) and matches the risk_classification band. • Every code in fraud_signals (FS-NNN) has at least one evidence_refs entry (APIM request ID or knowledge-base chunk ID). • public_records_check_status is set to COMPLETED, CACHED or UNAVAILABLE, and assessed_at is stamped. • The block validates against ClaimCaseOutput v1.0 and is returned within 12 tool calls or 120 seconds. If the budget runs out, the result is INDETERMINATE with the reason recorded.
Authority Boundaries	**Read-only everywhere.** It may read the claim case fields the controller passes in (extracted_claim, policy_cross_reference), public-records responses, fraud indicator guidance and 36-month claim history. **It may not:** write to the policy database, the claim case record, the settlement system or payments; change claim status; approve, deny or reserve a claim; contact the insured, broker or any third party; change thresholds or scoring weights; or call the Settlement Specialist or any other agent. Its only output is the fraud_assessment block returned to the Workflow Controller, which alone persists it. This is enforced technically: uami-claims-fraud-risk holds no write role (Section 1, Table 4a), APIM exposes GET operations only, and the SQL role grants SELECT on one view.
Tool Access Scope	• public_records_lookup (OpenAPI tool) → apim-apex-integration-prod /public-records/v1: GET entity-search, property-ownership, liens-judgments, litigation. Entra token from uami-claims-fraud-risk; 8-second timeout; APIM retry and circuit breaker (Section 4, FN-001). • Foundry IQ knowledge base kb-claims-fraud-indicators (Apex SIU red-flag guidelines and scoring rubric) on srch-apex-claims-prod, connected only to proj-apex-claims-processing. • get_claim_history (Azure Functions tool) → sqldb-apex-policy view vw_claim_history_36m, SELECT only. • score_fraud_signals (Azure Functions tool, deterministic): turns signal codes into a calibrated score using versioned weights (fraud-weights v2026.10). • **Not granted:** web search, code interpreter, email or Teams, file write, or any Legal or Finance resource.
Asynchronous	**Condition:** fraud_risk_score ≥ 0.85 (HIGH, ESC-002), or risk_classification = INDETERMINATE after the

Pillar	Learner Response
Escalation Checkpoint	fallback (ESC-005). **Mechanism:** the Workflow Controller writes a checkpoint (workflow state plus fraud_assessment) to Cosmos DB container workflow-checkpoints, publishes a message to Service Bus queue sbq-siu-fraud-review (HIGH) or sbq-external-verification-hold (INDETERMINATE) with CorrelationId = correlation_id, and suspends this claim's workflow. Other claims keep flowing. **Destination:** the SIU investigator queue (owner: SIU Manager), with a target response of 2 business days. INDETERMINATE claims go to the assigned adjuster after 4 hours. **Resume rule:** the workflow continues only when an authenticated reviewer decision event arrives. CLEAR hands off to Settlement with the reviewer ID recorded; CONFIRM closes the claim to SIU referral. The Settlement Specialist is never invoked while the checkpoint is open.

8. Grounding and Project Alignment Check

Agent	Grounding Source	Project Containing Source	Agent Project	Aligned?
Claim Intake / Extraction Agent	• claims-inbox on stapexclaimsintake • di-apex-claims-prod (model apex-acord1-property-loss-v3) • Foundry IQ kb-claims-intake-guide	proj-apex-claims-processing	proj-apex-claims-processing	**Yes**
Policy Cross-Reference Agent	• sqldb-apex-policy view vw_policy_coverage • Foundry IQ kb-claims-policy-wordings	proj-apex-claims-processing	proj-apex-claims-processing	**Yes**
Fraud and Risk Specialist	• Public-records API via apim-apex-integration-prod • Foundry IQ kb-claims-fraud-indicators • sqldb-apex-policy view vw_claim_history_36m	proj-apex-claims-processing	proj-apex-claims-processing	**Yes**
Settlement Specialist	• Foundry IQ kb-claims-settlement-templates • calculate_settlement tool	proj-apex-claims-processing	proj-apex-claims-processing	**Yes**

All four Foundry IQ knowledge bases are indexes on `srch-apex-claims-prod`, which is reached only through Project-scoped connections in `proj-apex-claims-processing` (Section 1, item 2). No Claims agent is grounded on a hub-level connection or on `srch-apex-legal-prod`, and the Legal compliance agent has no connection to any source in this table.

Submission Checklist

☑ Task agent trigger references a claim document ingestion event.

☑ Typed inputs include named fields and data types.

☑ JSON output schema can feed a downstream settlement system.

☑ At least two escalation conditions include specific thresholds.

☑ Fraud risk agent has goal state, definition of done, authority boundaries, tool access scope, and escalation checkpoint.

☑ Grounding sources align with the Project boundaries defined in Section 1.